

SPACECE INDIA FOUNDATION

Home Visit Observation (HAALS) Pipeline & Dashboard

Implementation Plan — Form to Dashboard to Report

Version 1.0

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Companion to: Teacher Training & Fellowship Portal — Complete Workflow (Section 7.2, HAALS) and the Home Visit Observation Form Blueprint (0–6 and 6–9 years).

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0. Purpose & Scope

This document lays out how data captured through the Home Visit Observation Google Form (0–6 and 6–9 year blueprints) gets pulled into the Teacher Training & Fellowship Portal, displayed on Fellow and Mentor dashboards, and turned into an auto-generated report for both roles.

This is an implementation plan, not a fresh product design — every piece below reuses a pattern that already exists and is live elsewhere in the portal (the approval mapping chain, the Fellow Activities review queue, and the Child Feedback AI structuring flow). No new infrastructure, login system, or app is introduced.

How to read this document:

- Section 1 — the seven-layer architecture, end to end, in one picture.
- Section 2 — the source data: what the Form actually captures.
- Section 3 — the database shape (VisitObservation) that everything else is built on.
- Section 4 — how form responses get from Google Sheets into the portal's database.
- Section 5 — what gets calculated, and how it differs for a Fellow vs. a Mentor.
- Section 6 — the two dashboard views, field by field.
- Section 7 — how the written report is generated and delivered.
- Section 8 — build order and phasing.
- Section 9 — open items and decisions needed before build starts.

1. Architecture Overview — Seven Layers

The system is best understood as seven layers, each one feeding the next. Layers 1–4 get the data ready; Layer 5 displays it; Layers 6–7 turn it into something a person acts on.

Layer	What it does	Built how
1. Collect	Fellow submits the Home Visit form during a visit.	Google Form — already built, no engineering needed.
2. Sync	New form responses are copied from the linked Google Sheet into the portal's own database.	Apps Script trigger on form-submit → webhook, or a scheduled job polling the Sheets API.
3. Organize	Raw answers are mapped into a structured record (which child, which Fellow, which activity, what score).	Parsing logic in the sync job, written into the VisitObservation collection (Section 3).
4. Calculate	Aggregate numbers are computed: visit completion rate, average milestone score per domain, participation rate, common challenges.	A scheduled aggregation job (e.g. nightly) or computed on read for smaller data volumes.
5. Display	Fellow dashboard shows their own numbers; Mentor dashboard shows all their Fellows' numbers rolled up.	New “Home Visits” tab on each existing dashboard, calling a read API.
6. Report	Numbers + the Fellow's written notes are turned into a short, plain-language summary.	Scheduled or on-demand call to an LLM, same pattern as the live Child Feedback (AI) module.

Layer	What it does	Built how
7. Deliver	The report is saved and shown on both dashboards, with an optional notification.	Stored report record + existing Notifications module (in-app / WhatsApp).

Layers 1–5 alone already produce a usable, working dashboard. Layers 6–7 (the written report) can ship as a second milestone without blocking the first — see Section 8.

2. Source Data — What the Form Captures

Two versions of the form exist, sharing the same structure: 0–6 years (6 domains, including Nutrition) and 6–9 years (5 domains, mapped to Curriculum Levels 13–18). Both follow the same seven-section shape.

Section	Captures	Why it matters downstream
1. Visit details	Date, Facilitator, Village/Area, Child, Age group, Program enrolled.	Drives every filter and rollup (by Fellow, by Centre, by Program).
2. Visit status	Is child present, is caregiver available, is child willing to participate.	If not “Yes,” the visit ends early — this itself is a metric (attempted vs. completed visits), not something to discard.
3. Home environment	Space adequacy, materials available, whether household items substitute for toys.	Feeds the “Home environment rating” trend on the Mentor view.
4. Developmental observation	1–2 activities logged, each with domain, engagement level, attempt/completion, support needed, and a 0–5 milestone score.	This is the core analytics field — milestone score by domain is what powers the milestone rings on both dashboards.
5. Parent engagement	Did the caregiver observe/participate, can they repeat it, what helped, what challenges came up, follow-up on last week's activity.	Powers “parent participation rate” and the “common challenges” list.
6. Follow-up	Recommended next action (repeat, progress milestone, refer for screening, parent counselling, etc.).	Feeds the follow-ups-pending count and the flagged-children logic.
7. End of visit	Child participation, parent cooperation, home environment — each rated 1–5 — plus facilitator remarks and photos.	Remarks feed the AI report; ratings feed trend charts.

3. Data Model — VisitObservation

The existing portal has a generic “Session” structure used by Parent Capacity Building, Home Visits, and Library. It is too thin for this form's nested activities, checkbox sets, and ratings, so a dedicated collection is recommended, linked into the same Program → Centre → Mentor → Fellow → Family → Child mapping already used everywhere else in the portal.

Field group	Fields	Notes
Visit meta	visitDate, facilitatorId (→ Fellow), village, childId, childName, ageGroup, program	program maps directly onto the existing PTP / School Readiness / FLN / Library enum.
Visit status	childPresent, caregiverAvailable, childWillingness	Store the record even on an incomplete visit — completion rate depends on it.
Home environment	spaceAdequate, materialsAvailable[], householdItemsUsable	
Activities (array)	activityName, milestoneSource, domain[], engagementLevel, attempted, completed, supportNeeded, milestoneStatus (0/1–2/3/4/5)	One entry per activity logged in the visit (1–2 typically). milestoneStatus is the single most valuable field for scoring.
Parent engagement	caregiverObserved, caregiverParticipated, canRepeatAtHome, helpFactors[], challenges[], isFollowUp, didLastWeekActivity, lastWeekCompletionCount, lastWeekDifficulties, homeActivitiesAssigned	
Follow-up	recommendedAction (enum)	
Closing	childParticipationRating, parentCooperationRating, homeEnvironmentRating (1–5 each), remarks, photos[]	remarks is the primary free-text input to the AI report in Section 7.

4. Sync Pipeline — Form to Database

4.1 Recommended approach

An Apps Script bound to the Form's response Sheet, triggered on every submit, sends the new row directly to a webhook endpoint on the portal backend (e.g. POST /api/haals/visits). This is close to real time and avoids a separate polling job.

4.2 Alternative approach

A scheduled backend job (e.g. every few minutes) reads new rows from the Sheet via the Google Sheets API and upserts them. Slightly simpler to set up, with a small delay before data appears.

4.3 Parsing notes

- Activities repeat as numbered question blocks (“Name of the Activity - 1”, “Name of the Activity - 2”) — the parser loops these into the activities[] array rather than storing them as flat fields.
- The 0–6 and 6–9 forms use slightly different domain sets (6 vs. 5 domains, Nutrition only on the younger form) — the parser should branch on which form version submitted, but both write into the same VisitObservation shape.
- Skipped sections (child not present / caregiver unavailable) mean most fields are null — this is expected and should not cause the sync to fail.

5. Aggregation & Metrics

The dashboards and reports don't run new logic per se — they read pre-calculated numbers from an aggregation layer that groups VisitObservation records by Fellow (for the Fellow view) or by Mentor's full Fellow list (for the Mentor view).

Metric	Fellow view (own data)	Mentor view (rolled up)
Visit completion	Completed vs. scheduled, with reasons for incomplete visits	Same, per Fellow, side by side
Milestone score	Average 0–5 score per domain, trended over time	Average per domain per Fellow, plus Centre-wide average
Engagement	Distribution across Highly engaged / Moderate / Low / Needs support / Not interested	Same, aggregated across all Fellows
Parent participation	% of visits with caregiver participation	Same, plus trend by Fellow
Common challenges	Top challenge tags this period	Top challenge tags Centre-wide
Flagged children	Not applicable at individual level (see Mentor view)	Any child with ≥ 2 visits showing “Not Attempted” or low engagement on the same domain — same flagging rule already live in Child Feedback (AI)

6. Dashboard Specification

6.1 Fellow view — “Home Visits” tab

- KPI cards: visits completed / scheduled, average milestone score, parent participation rate, follow-ups pending.
- Milestone score by domain, shown as a small progress indicator per domain.
- Recent visits list: child, date, activity, engagement level, status.
- Common challenges this period, as a simple tag list.

6.2 Mentor view — “Home Visits” tab

- KPI cards: Centre-wide visit completion, average milestone score, count of flagged children.
- Per-Fellow comparison table: visit completion rate and average milestone score, side by side — surfaces who may need coaching support.
- Children needing attention: auto-flagged list with the Fellow assigned and the reason flagged.

6.3 Implementation note

Both views are new tabs added to the existing Fellow and Mentor dashboards (the same components already rendering Fellow Activities and Curriculum Management), reading from the two endpoints in Section 5. No new authentication, navigation shell, or design system is required.

7. Report Generation

7.1 Inputs

The aggregated metrics from Section 5, plus the free-text remarks field from each visit in the period, are passed to an LLM with an instruction to summarize in plain language and flag concerns — the

same input/output pattern already live in the Child Feedback (AI) module (teacher writes free text, AI structures it, Admin/Mentor reviews).

7.2 Output structure

- Fellow report: strengths observed, areas needing support, suggested focus for next visits, summary of parent engagement.
- Mentor report: same structure, rolled up across all assigned Fellows, plus a short note on which Fellows may need coaching support based on the comparison table.

7.3 Cadence

Recommended: generate automatically on a weekly or monthly schedule, with an on-demand “Generate Report” option available at any time. Each generated report is stored (not just emailed once) so it remains viewable on the dashboard later.

7.4 Delivery

Reports appear directly on the relevant dashboard, with an optional notification through the portal's existing Notifications module (in-app, email, or WhatsApp, matching the channel the person has already configured in their profile).

8. Build Order & Phasing

Phase	Scope	Outcome
Phase 1	VisitObservation model, sync pipeline (Section 4), basic Fellow + Mentor list views	Raw visit data is visible and searchable in the portal
Phase 2	Aggregation layer (Section 5), full dashboard views with KPI cards and milestone indicators (Section 6)	A working, live dashboard for both roles
Phase 3	AI report generation and delivery (Section 7)	Automated written reports for Fellow and Mentor
Phase 4	Reminders (visit due, follow-up due) and cross-program reporting, matching existing PCB/HAALS reminder patterns	Fully automated, self-maintaining module

9. Open Items & Decisions Needed

- Domain grouping: the 0–6 form has 6 domains (including Nutrition); the 6–9 form has 5. Confirm whether Nutrition should be its own separate metric rather than folded into a shared domain view.
- Reminder channel and timing thresholds (in-app, SMS, WhatsApp, email; how many hours/days before due) — same open item already flagged for PCB and HAALS in the portal workflow document.
- Report cadence — confirm weekly vs. monthly as the default automatic schedule.
- Flagging threshold — confirm “≥ 2 visits” is the right sensitivity for the children-needing-attention flag, or whether it should vary by domain.

Appendix — Source References

- Teacher Training & Fellowship Portal — Complete Workflow, v1.0 (Section 7.2, HAALS)

- Teacher Training Portal — Complete Platform Workflow, v1.0 (Child Feedback AI pattern, Section 8)
- Fellowship Portal — Roles, Registration, Approval & Dashboard Workflow, v1.0
- Home Visit Observation Form Blueprint — Revised (0–6 years and 6–9 years)